Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

On heating aqueous solution of benzene diazonium chloride, which of the following is formed?

(A) benzene

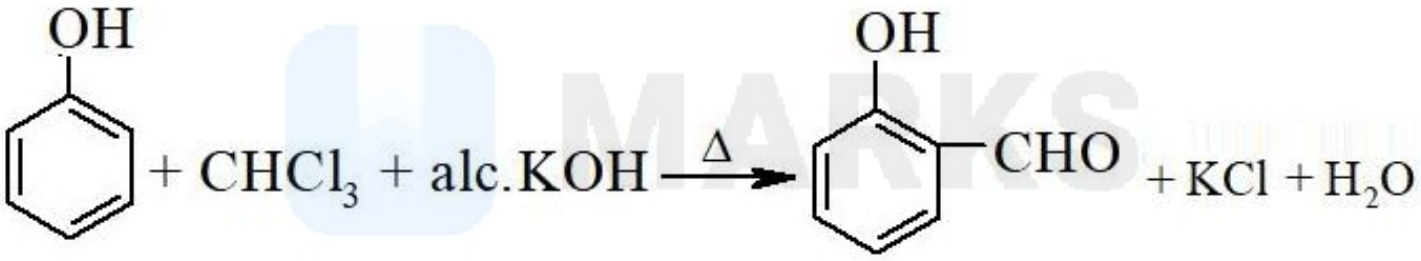
(B) chlorobenzene

(C) phenol

(D) aniline

Q.2

For the reaction



The correct statement regarding above reaction is

(A) The reaction completes in more than one step

(B) Dichloro carbene is an electrophile which attack on benzene ring

(C) 

intermediate is formed during reaction

(D) A & B both are correct

Q.3

The reaction of phenol with benzoyl chloride to give phenyl benzoate is known as:

(A) Claisen reaction

(B) Schotten-Baumann reaction

@IITJEE_Advanced

(C) Reimer-Tiemann reaction

(D) Gatterman-Koch reaction

Q.4

Phenol on heating with $CHCl_3$ and $NaOH$ gives salicylaldehyde. The reaction is called:

(A) Reimer - Tiemann reaction

(B) Claisen reaction

(C) Cannizzaro's reaction

(D) Hell - Volhard - Zelinsky reaction

Q.5

What amount of bromine will be required to convert $2\ g$ of phenol into $2, 4, 6$ -tribromophenol?

(A) 4.0

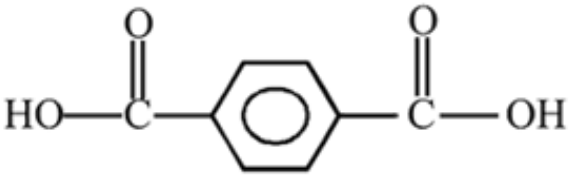
(B) 6.0

(C) 10.22

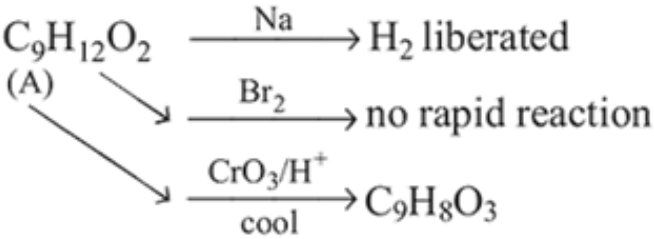
(D) 20.44

Q.6

An optically active alcohol of formula $C_9H_{12}O_2$ produced the following compound when refluxed with $KMnO_4$



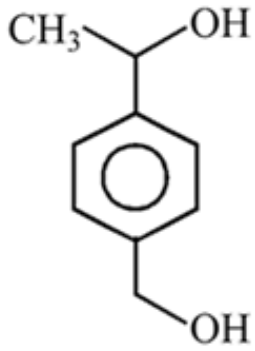
The original compound showed these properties also:



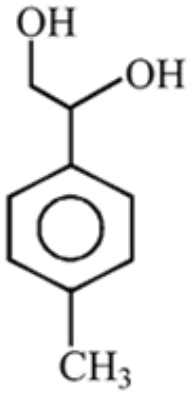
What is structure of (A)?

@IITJEE_Advanced

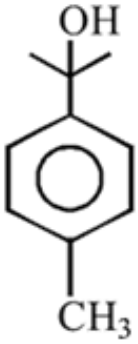
(A)



(B)



(C)



(D) both (a) and (b)

Q.7

Which of the following reactions will not result in the formation of anisole?

(A) Phenol + dimethyl sulphate in presence of a base

(B) Sodium phenoxide is treated with methyl iodide

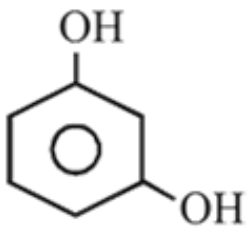
(C) Reaction of diazomethane with phenol

(D) Reaction of methylmagnesium iodide with phenol

Q.8

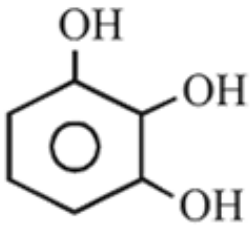
Which of the following compound can react with hydroxylamine?

(A)

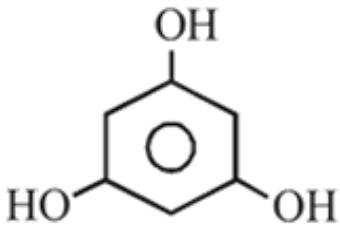


@IITJEE_Advanced

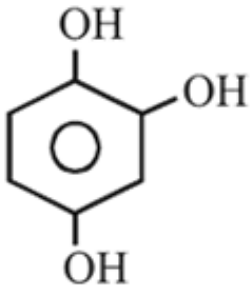
(B)



(C)



(D)



Q.9

In order to get 2 -hydroxybenzaldehyde from phenol, which of the following reagents is required?

(A) $(CH_3CO)_2O, H_2SO_4$

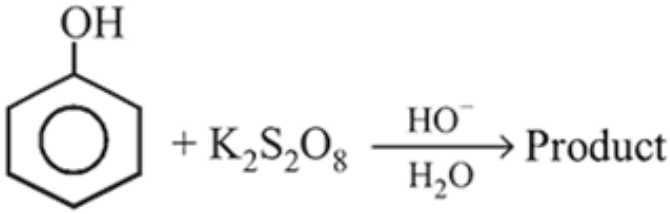
(B) $CHCl_3/NaOH$

(C) $CO_2, NaOH$

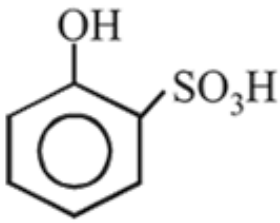
(D) $CCl_4/NaOH$

Q.10

Identify the nature of product in the following reaction

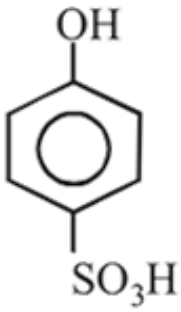


(A)

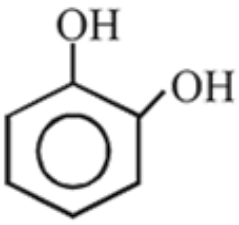


@IITJEE_Advanced

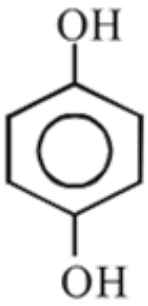
(B)



(C)



(D)



Q.11

Phenol can be distinguished from alcohol with

(A) Tollens reagent

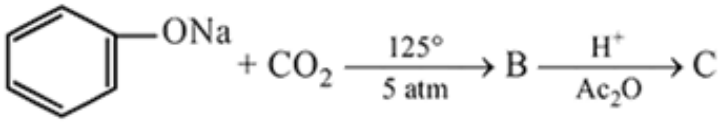
(B) Schiff's base

(C) Neutral $FeCl_3$

(D) HCl

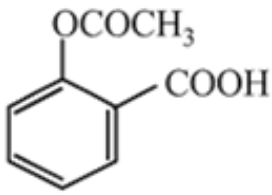
Q.12

Sodium phenoxide when heated with CO_2 under pressure at $125^\circ C$ yields a product which on acetylation produces C



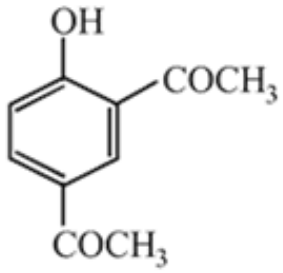
The major product C would be

(A)

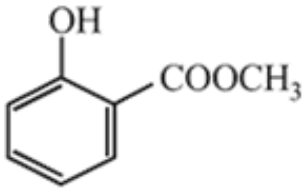


@IITJEE_Advanced

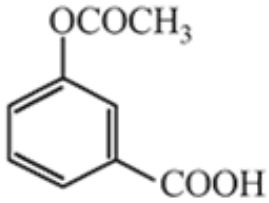
(B)



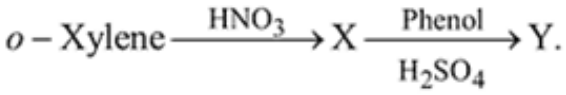
(C)



(D)



Q.13



The product Y is

(A) Phthalic acid

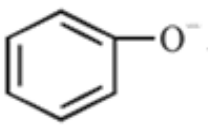
(B) Isophthalic acid

(C) Phenolphthalein

(D) *o* -Hydroxysulphonic acid

Q.14

Which one of the following substituents at paraposition is most effective in stabilizing the phenoxide



ion?

(A) $-CH_3$

(B) $-OCH_3$

(C) $-COCH_3$

@IITJEE_Advanced

(D) $-CH_2OH$

Q.15

1-Phenylethanol can be prepared by reaction of benzaldehyde with

(A) methyl bromide

(B) ethyl iodide and magnesium

(C) methyl bromide and aluminium bromide

(D) methyl iodide and magnesium

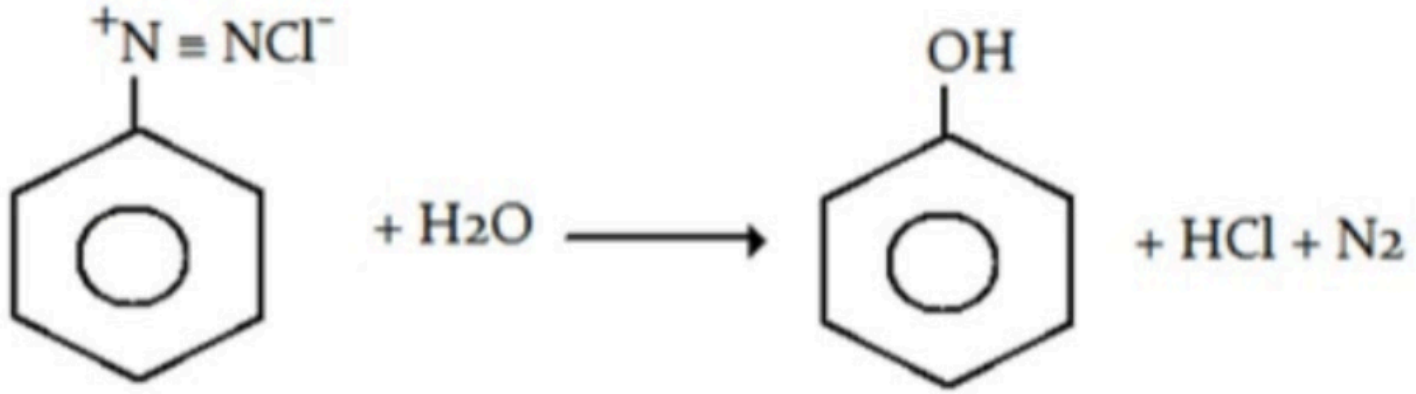
Answers & Solutions

Q.1 Answer:

phenol

Solution:

Upon warming with water, these diazonium salts finally hydrolyze to phenols.

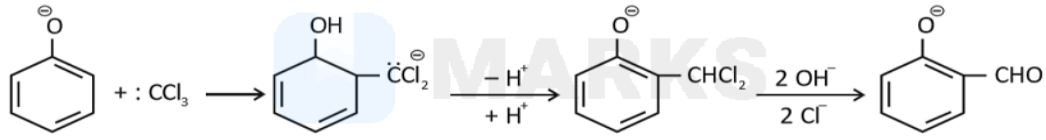


Q.2 Answer:

A & B both are correct

Solution:

$$CHCl_3 + KOH \xrightarrow{\Delta} :CCl_2 + KCl + H_2O$$

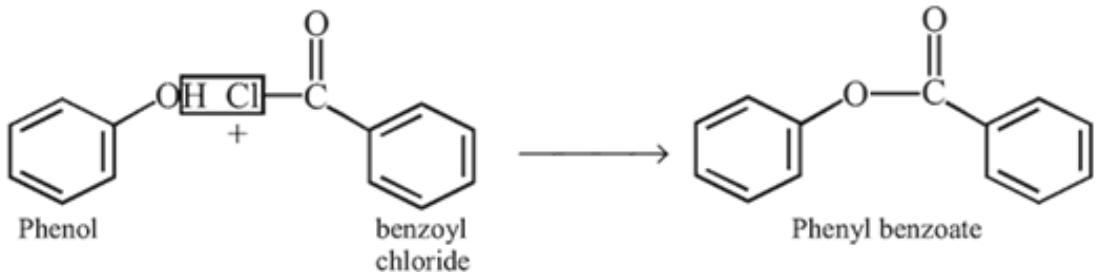


Q.3 Answer:

Schotten-Baumann reaction

Solution:

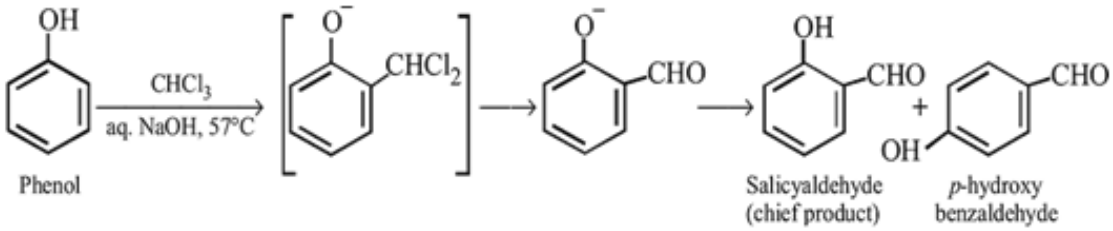
@IITJEE_Advanced



Q.4 Answer:

Reimer - Tiemann reaction

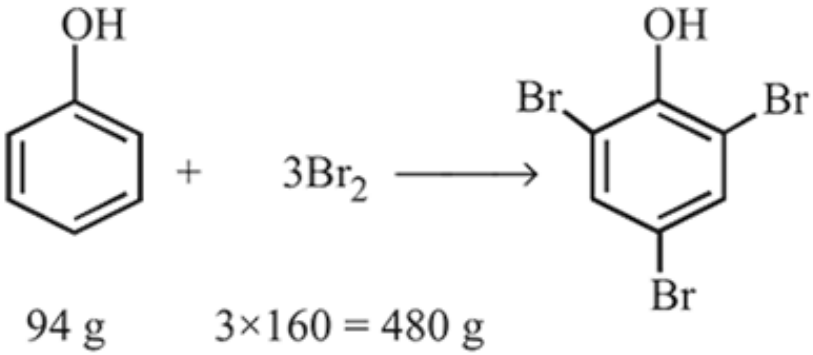
Solution:



Q.5 Answer:

10.22

Solution:



Thus 94 g **of phenol require** $Br_2 = 480$ g

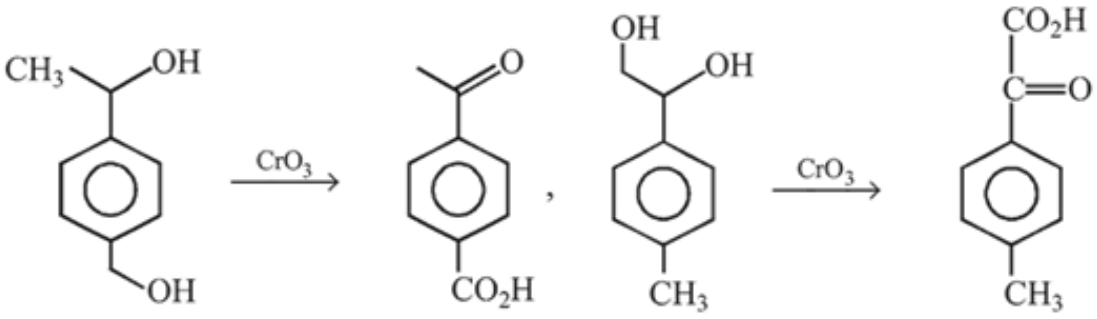
2 g **of phenol require** $Br_2 = \frac{480}{94} \times 2 = 10.22$ g

Q.6 Answer:

both (a) and (b)

Solution:

@IITJEE_Advanced



Q.7 Answer:

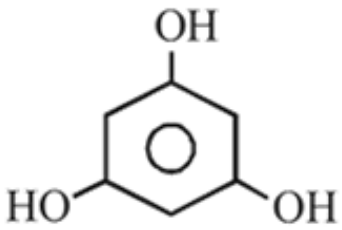
Reaction of methylmagnesium iodide with phenol

Solution:

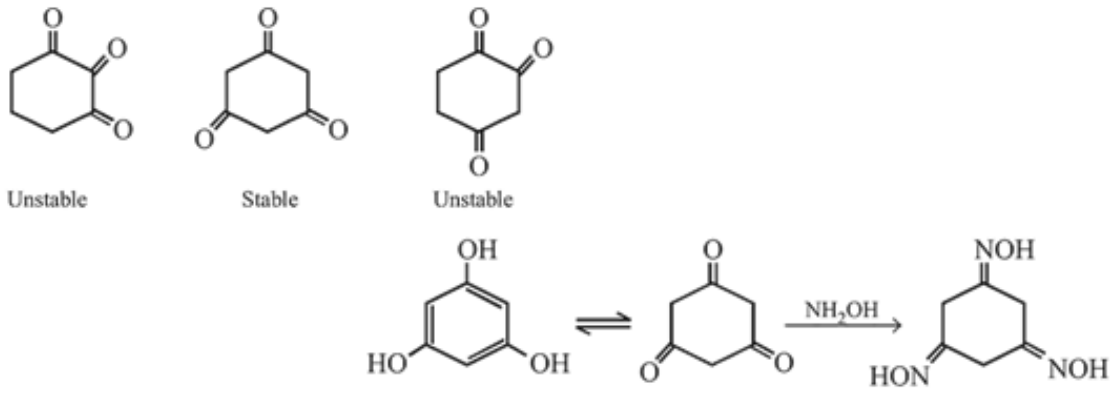
Phenol has active (acidic) hydrogen so it reacts with CH_3MgI to give CH_4, and not anisole

$$C_6H_5OH + CH_3MgI \longrightarrow CH_4 + C_6H_5OMgI$$

Q.8 Answer:



Solution:

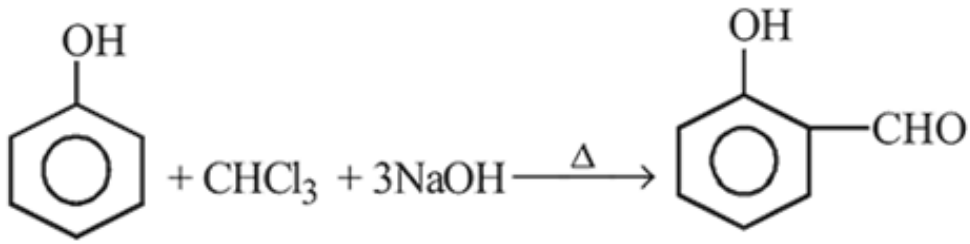


Q.9 Answer:

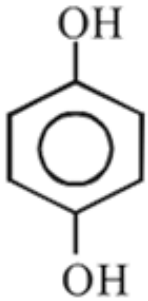
$CHCl_3/NaOH$

Solution:

@IITJEE_Advanced



Q.10 Answer:



Solution:

Oxidation of phenol by $K_2 S_2O_8$ is an example of Elbs persulphate oxidation.

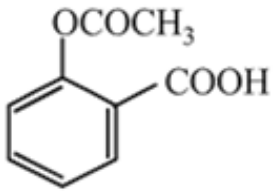
Q.11 Answer:

Neutral $FeCl_3$

Solution:

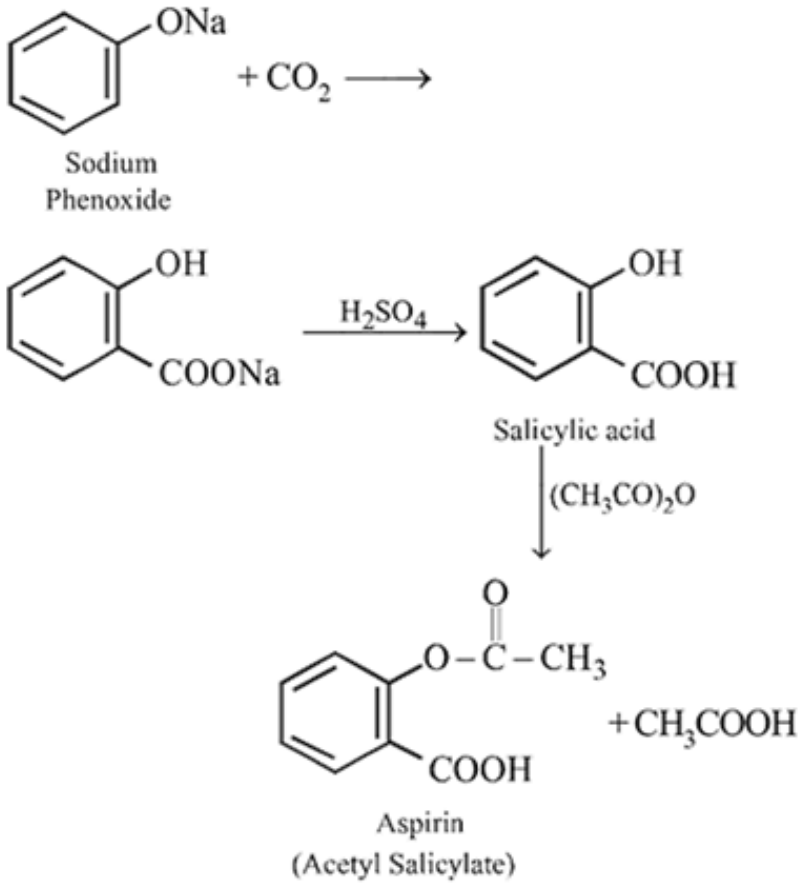
Due to the formation of complex $[Fe(OC_6H_5)_6]$ Phenol gives violet colour with neutral $FeCl_3$

Q.12 Answer:



Solution:

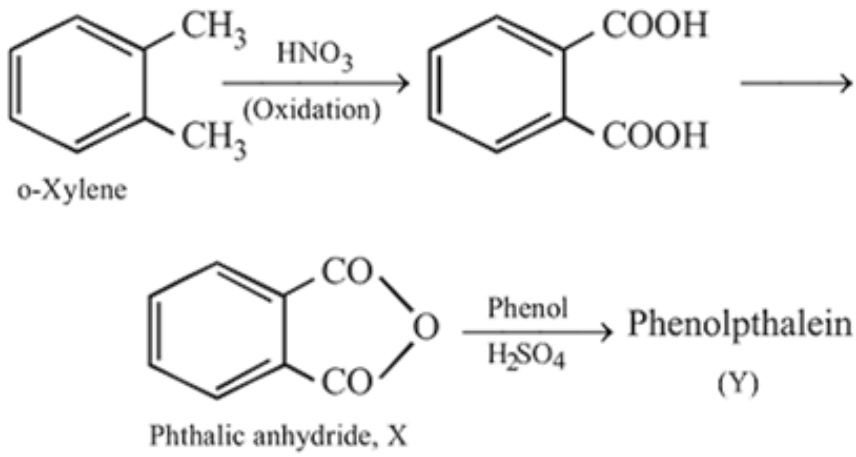
@IITJEE_Advanced



Q.13 Answer:

Phenolphthalein

Solution:



Q.14 Answer:

$-COCH_3$

Solution:

Electron withdrawing group stabilises the benzene ring due to delocalisation of charge.

CH_3 **and** $-CH_2OH$ **are electron donating group and hence decrease the stability of benzene ring**

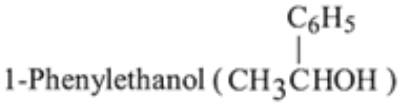
@IITJEE_Advanced

$-OCH_3$ is weaker electron withdrawing group than $-COCH_3$. Hence $-COCH_3$ group more stabilize the phenoxide ion at p -position.

Q.15 Answer:

methyl iodide and magnesium

Solution:



is a 2° alcohol and can be prepared by the reaction of benzaldehyde with Grignard reagent CH_3MgI.

@IITJEE_Advanced